

Name: J. Renuka Devi

Reg.no: 192573007

Course code: DSA0602

Course name: Data Handling and Visualization

CLASS ACTIVITY 15.07.2026

Python Code

1. Smart City Air Quality Monitoring:

```
import pandas as pd
import matplotlib.pyplot as plt

air = pd.DataFrame({
    "Temperature": [25, 28, 30, 32, 35],
    "PM2.5": [40, 55, 70, 90, 110]
})

plt.scatter(air["Temperature"], air["PM2.5"],
            c=air["PM2.5"], cmap="Blues", s=150)

plt.colorbar(label="PM2.5")

plt.title("PM2.5 Hotspots (Sequential Color Scale)")

plt.xlabel("Temperature (°C)")

plt.ylabel("PM2.5")

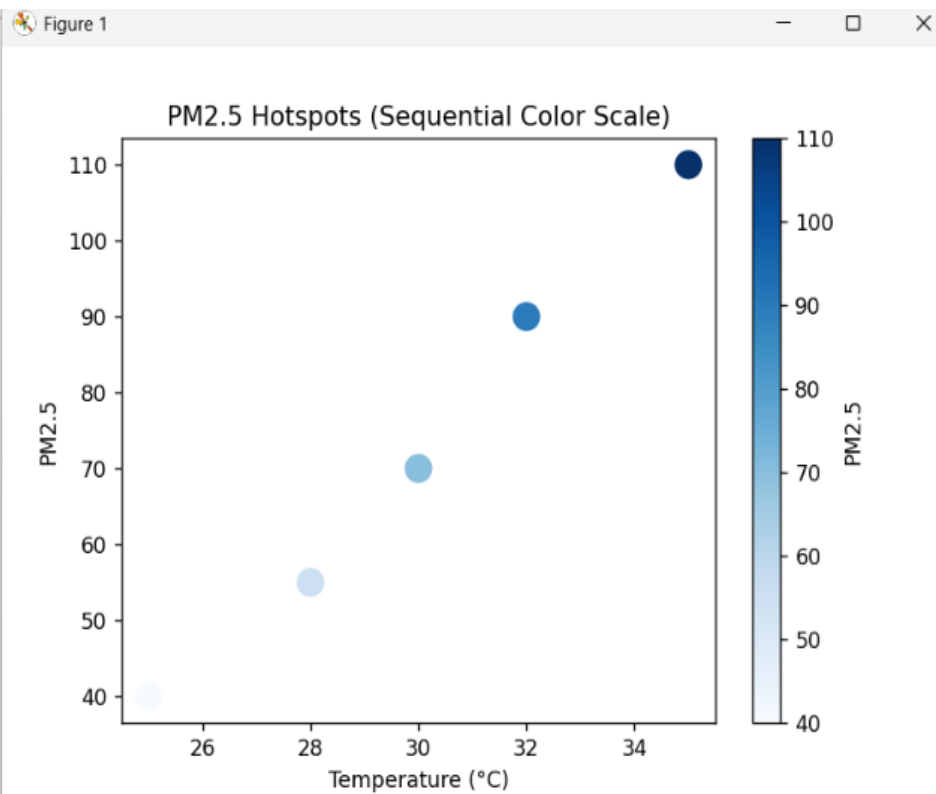
plt.show()
```

```
import pandas as pd
import matplotlib.pyplot as plt

air = pd.DataFrame({
    "Temperature": [25, 28, 30, 32, 35],
    "PM2.5": [40, 55, 70, 90, 110]
})

plt.scatter(air["Temperature"], air["PM2.5"],
            c=air["PM2.5"], cmap="Blues", s=150)

plt.colorbar(label="PM2.5")
plt.title("PM2.5 Hotspots (Sequential Color Scale)")
plt.xlabel("Temperature (°C)")
plt.ylabel("PM2.5")
plt.show()
```



2. Earthquake Risk Analysis:

```
import pandas as pd

import matplotlib.pyplot as plt

air = pd.DataFrame({
    "Temperature": [25, 28, 30, 32, 35],
    "PM2.5": [40, 55, 70, 90, 110]
})

plt.scatter(air["Temperature"], air["PM2.5"],
            c=air["PM2.5"], cmap="coolwarm", s=150)

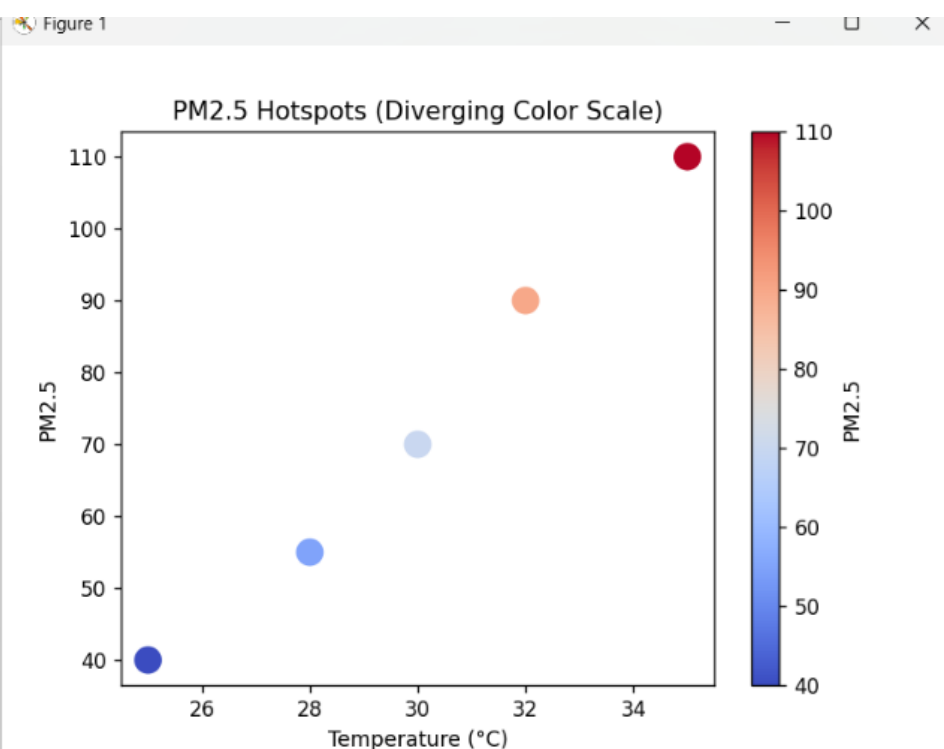
plt.colorbar(label="PM2.5")
plt.title("PM2.5 Hotspots (Diverging Color Scale)")
plt.xlabel("Temperature (°C)")
plt.ylabel("PM2.5")
plt.show()
```

```
import pandas as pd
import matplotlib.pyplot as plt

air = pd.DataFrame({
    "Temperature": [25, 28, 30, 32, 35],
    "PM2.5": [40, 55, 70, 90, 110]
})

plt.scatter(air["Temperature"], air["PM2.5"],
            c=air["PM2.5"], cmap="coolwarm", s=150)

plt.colorbar(label="PM2.5")
plt.title("PM2.5 Hotspots (Diverging Color Scale)")
plt.xlabel("Temperature (°C)")
plt.ylabel("PM2.5")
plt.show()
```



3.Wind Direction Analysis for Airport Operations:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
wind = pd.DataFrame({
    "Direction":["N","NE","E","SE","S","SW","W","NW"],
    "Speed":[18,25,15,10,8,12,20,16]
})
```

```
# Cartesian Plot
```

```
plt.bar(wind["Direction"], wind["Speed"], color="skyblue")
plt.title("Wind Speed by Direction")
plt.xlabel("Direction")
plt.ylabel("Speed")
plt.show()
```

```
# Polar Plot
```

```
angles=np.linspace(0,2*np.pi,8,endpoint=False)
```

```
ax=plt.subplot(111,projection="polar")
ax.bar(angles,wind["Speed"],width=0.5,
       color=plt.cm.Blues(np.array(wind["Speed"])/25))
plt.title("Wind Rose")
plt.show()
```

```
print("Best Visualization: Polar Plot (Wind Rose)")
```

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

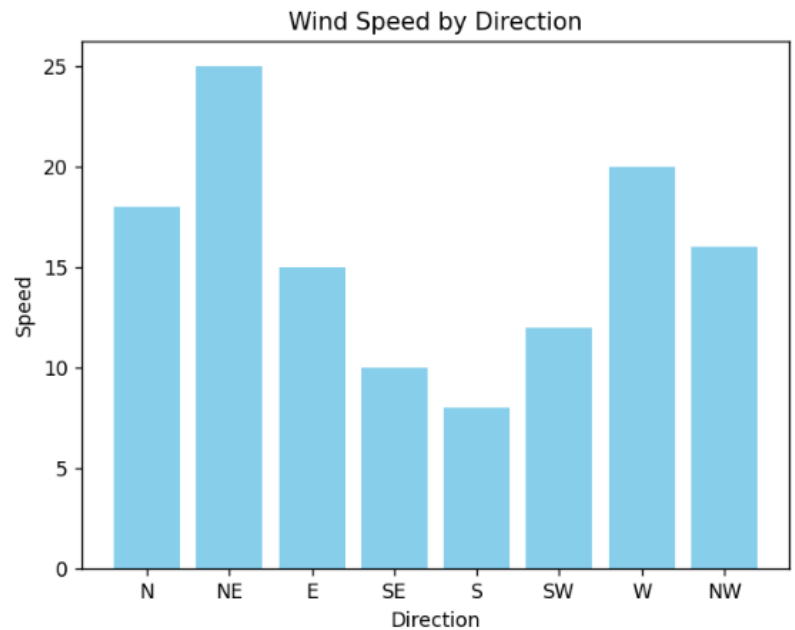
wind = pd.DataFrame({
    "Direction":["N","NE","E","SE","S","SW","W","NW"],
    "Speed": [18,25,15,10,8,12,20,16]
})

# Cartesian Plot
plt.bar(wind["Direction"], wind["Speed"], color="skyblue")
plt.title("Wind Speed by Direction")
plt.xlabel("Direction")
plt.ylabel("Speed")
plt.show()

# Polar Plot
angles=np.linspace(0,2*np.pi,8,endpoint=False)

ax=plt.subplot(111,projection="polar")
ax.bar(angles,wind["Speed"],width=0.5,
       color=plt.cm.Blues(np.array(wind["Speed"])/25))
plt.title("Wind Rose")
plt.show()

print("Best Visualization: Polar Plot (Wind Rose)")
```



4 – Hospital Patient Monitoring Dashboard

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
patient = pd.DataFrame({
    "Patient":["P1","P2","P3","P4","P5"],
    "HeartRate":[72,95,120,65,140]
})
```

```
colors=["green","green","red","green","red"]
```

```
plt.bar(patient["Patient"],patient["HeartRate"],color=colors)
```

```
plt.title("Hospital Patient Monitoring")
plt.xlabel("Patient")
```

```
plt.ylabel("Heart Rate")
plt.show()
```

```
print("Green = Normal")
print("Red = Critical")
```

```
import pandas as pd
import matplotlib.pyplot as plt

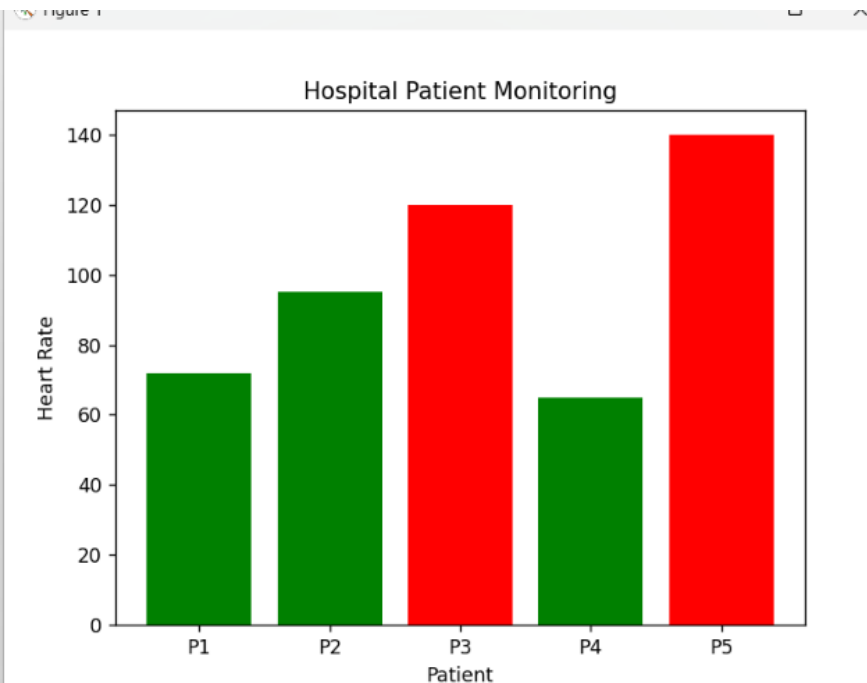
patient = pd.DataFrame({
    "Patient": ["P1", "P2", "P3", "P4", "P5"],
    "HeartRate": [72, 95, 120, 65, 140]
})

colors = ["green", "green", "red", "green", "red"]

plt.bar(patient["Patient"], patient["HeartRate"], color=colors)

plt.title("Hospital Patient Monitoring")
plt.xlabel("Patient")
plt.ylabel("Heart Rate")
plt.show()

print("Green = Normal")
print("Red = Critical")
```



5 – Global Climate Change Study

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
climate = pd.DataFrame({
    "Country": ["A", "B", "C", "D", "E"],
    "Anomaly": [-1.2, -0.5, 0.3, 0.8, 1.5]
})
```

```
# Palette 1
```

```
plt.imshow([climate["Anomaly"]], cmap="bwr", aspect="auto")
plt.colorbar()
plt.title("Blue-White-Red")
plt.xticks(range(5), climate["Country"])
plt.yticks([])
plt.show()
```

```
# Palette 2
```

```
plt.imshow([climate["Anomaly"]], cmap="coolwarm", aspect="auto")
plt.colorbar()
```

```

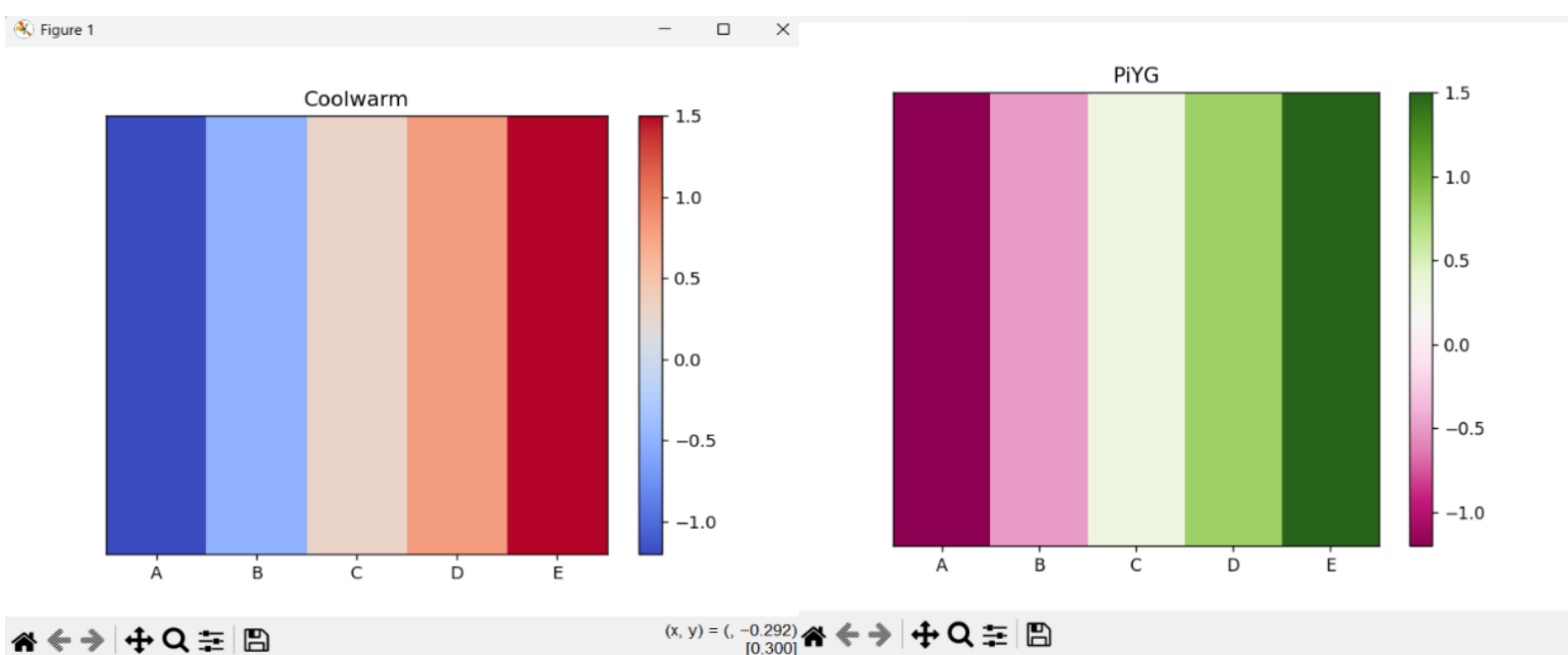
plt.title("Coolwarm")
plt.xticks(range(5),climate["Country"])
plt.yticks([])
plt.show()

# Palette 3
plt.imshow([climate["Anomaly"]],cmap="PiYG",aspect="auto")
plt.colorbar()
plt.title("PiYG")
plt.xticks(range(5),climate["Country"])
plt.yticks([])
plt.show()

print("Best Visualization: Blue-White-Red Palette")

```





6 – E-Commerce Sales Analytics

```
import pandas as pd
import matplotlib.pyplot as plt
sales = pd.DataFrame({
    "Category":["Electronics","Clothing","Furniture","Beauty","Sports"],
    "Sales":[85000,60000,45000,30000,50000]
})
plt.figure(figsize=(8,4))
plt.scatter(
    sales["Category"],
    sales["Sales"],
    c=sales["Sales"],
    cmap="viridis",
    s=300
)
plt.colorbar(label="Sales Volume")
plt.title("E-Commerce Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales")

plt.xticks(rotation=45)
plt.show()
print("Color represents sales volume and categories are distinguished using labels.")
print("Challenge: Too many categories may create confusing color combinations.")
```

```
import pandas as pd
import matplotlib.pyplot as plt

sales = pd.DataFrame({
    "Category": ["Electronics", "Clothing", "Furniture", "Beauty", "Sports"],
    "Sales": [85000, 60000, 45000, 30000, 50000]
})

plt.figure(figsize=(8,4))

plt.scatter(
    sales["Category"],
    sales["Sales"],
    c=sales["Sales"],
    cmap="viridis",
    s=300
)

plt.colorbar(label="Sales Volume")

plt.title("E-Commerce Sales Analysis")
plt.xlabel("Product Category")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.show()

print("Color represents sales volume and categories are distinguished using labels.")
print("Challenge: Too many categories may create confusing color combinations.")
```



7 – Satellite Image Vegetation Analysis

```
import pandas as pd
import matplotlib.pyplot as pl

ndvi = pd.DataFrame({
    "Area": ["Forest", "Grassland", "Water", "Barren", "Agriculture"],
    "NDVI": [0.85, 0.65, -0.8, -0.3, 0.55]
})

# Sequential Palette
plt.imshow(
    [ndvi["NDVI"]],
    cmap="Greens",
    aspect="auto"
)

plt.colorbar(label="NDVI Value")
plt.title("Sequential Palette - Vegetation Health")
plt.xticks(
    range(5),
    ndvi["Area"]
)

plt.yticks([])
plt.show()

# Diverging Palette
plt.imshow(
    [ndvi["NDVI"]],
    cmap="RdYlGn",
```

```

        aspect="auto"
    )
    plt.colorbar(label="NDVI Value")
    plt.title("Diverging Palette - Healthy vs Non Healthy Areas")
    plt.xticks(
        range(5),
        ndvi["Area"]
    )
    plt.yticks([])
    plt.show()
    print("Best Visualization: Diverging palette clearly separates vegetation and non-vegetation.")

```

```

import pandas as pd
import matplotlib.pyplot as plt

ndvi = pd.DataFrame({
    "Area": ["Forest", "Grassland", "Water", "Barren", "Agriculture"],
    "NDVI": [0.85, 0.65, -0.8, -0.3, 0.55]
})

# Sequential Palette
plt.imshow(
    [ndvi["NDVI"]],
    cmap="Greens",
    aspect="auto"
)

plt.colorbar(label="NDVI Value")

plt.title("Sequential Palette - Vegetation Health")

plt.xticks(
    range(5),
    ndvi["Area"]
)

plt.yticks([])

plt.show()

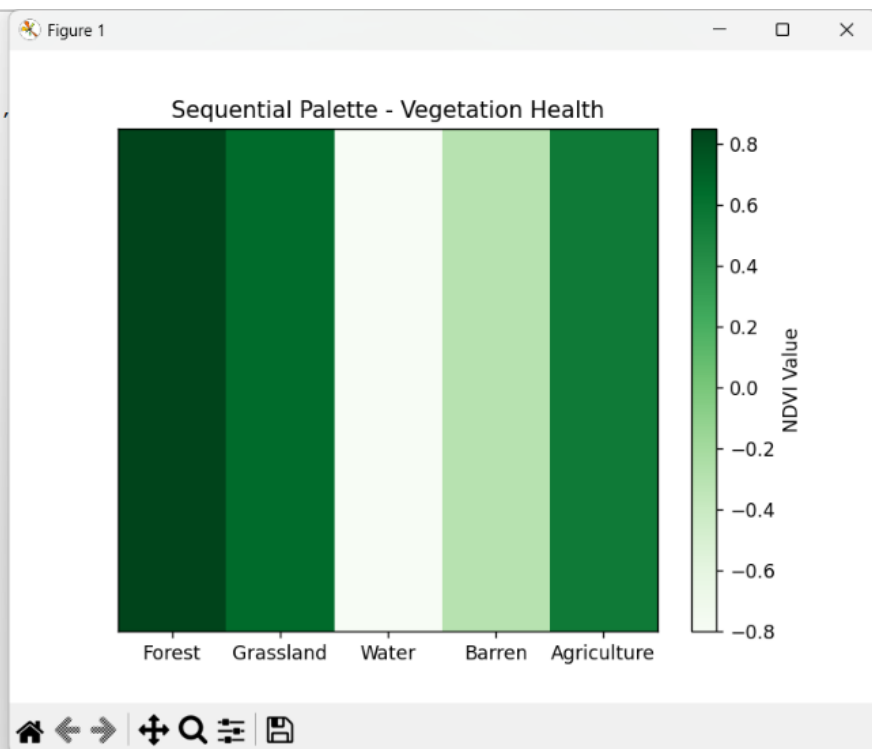
# Diverging Palette
plt.imshow(
    [ndvi["NDVI"]],
    cmap="RdYlGn",
    aspect="auto"
)

plt.colorbar(label="NDVI Value")

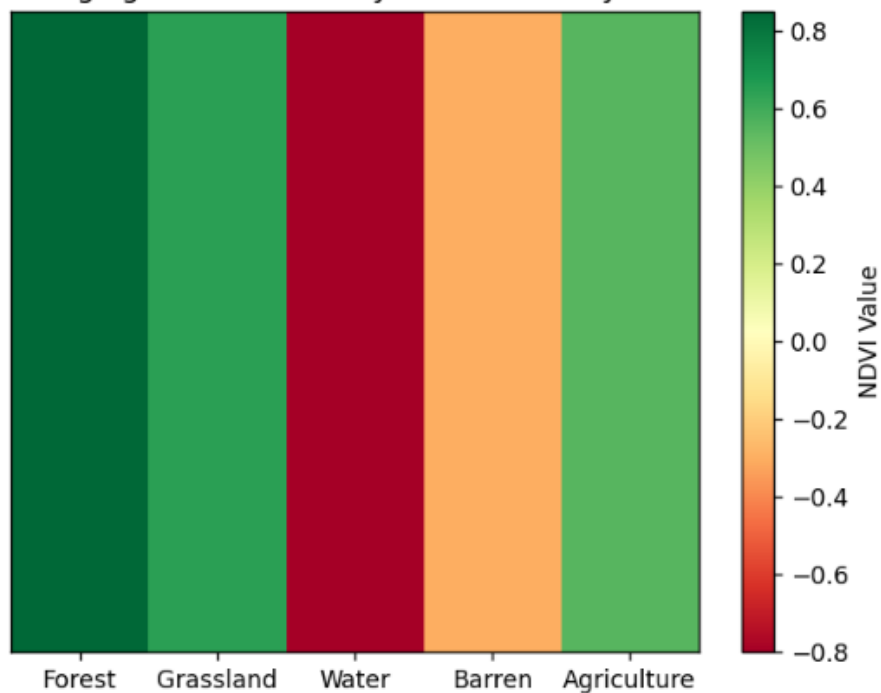
plt.title("Diverging Palette - Healthy vs Non Healthy Areas")

plt.xticks(
    range(5),
    ndvi["Area"]
)

```



Diverging Palette - Healthy vs Non Healthy Areas



(x, y) = (, -0.214)
[0.850]

8 – Cryptocurrency Market Volatility

```
import pandas as pd
import matplotlib.pyplot as plt

crypto = pd.DataFrame({
    "Day": [1, 2, 3, 4, 5],
    "Bitcoin": [20000, 25000, 30000, 35000, 40000],
    "Ethereum": [1500, 1800, 2200, 2500, 3000],
    "Solana": [40, 55, 70, 90, 120],
    "Cardano": [0.5, 0.7, 0.9, 1.1, 1.3]
})

# Cartesian Plot
plt.plot(
    crypto["Day"],
    crypto["Bitcoin"],
    label="Bitcoin"
)

plt.plot(
    crypto["Day"],
    crypto["Ethereum"],
    label="Ethereum"
```

```
)  
plt.plot(  
    crypto["Day"],  
    crypto["Solana"],  
    label="Solana"  
  
plt.plot(  
    crypto["Day"],  
    crypto["Cardano"],  
    label="Cardano"  
)  
plt.title("Cryptocurrency Prices - Cartesian Axis")  
plt.xlabel("Days")  
plt.ylabel("Price")  
plt.legend(  
plt.show(  
# Logarithmic Plot  
plt.plot(  
    crypto["Day"],  
    crypto["Bitcoin"],  
    label="Bitcoin"  
)  
plt.plot(  
    crypto["Day"],  
    crypto["Ethereum"],  
    label="Ethereum"  
)  
plt.plot(  
    crypto["Day"],  
    crypto["Solana"],  
    label="Solana"  
)  
plt.plot(  
    crypto["Day"],  
    crypto["Cardano"],  
    label="Cardano"  
  
plt.yscale("log")  
plt.title("Cryptocurrency Prices - Logarithmic Axis")  
plt.xlabel("Days")  
plt.ylabel("Log Price")
```

```
plt.legend()
plt.show()
print("Logarithmic axes help compare cryptocurrencies with different price ranges.")
print("Limitation: Log scales are harder for beginners to understand.")
```

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
crypto = pd.DataFrame({
    "Day": [1, 2, 3, 4, 5],
    "Bitcoin": [20000, 25000, 30000, 35000, 40000],
    "Ethereum": [1500, 1800, 2200, 2500, 3000],
    "Solana": [40, 55, 70, 90, 120],
    "Cardano": [0.5, 0.7, 0.9, 1.1, 1.3]
})
```

```
# Cartesian Plot
```

```
plt.plot(
    crypto["Day"],
    crypto["Bitcoin"],
    label="Bitcoin"
)
```

```
plt.plot(
    crypto["Day"],
    crypto["Ethereum"],
    label="Ethereum"
)
```

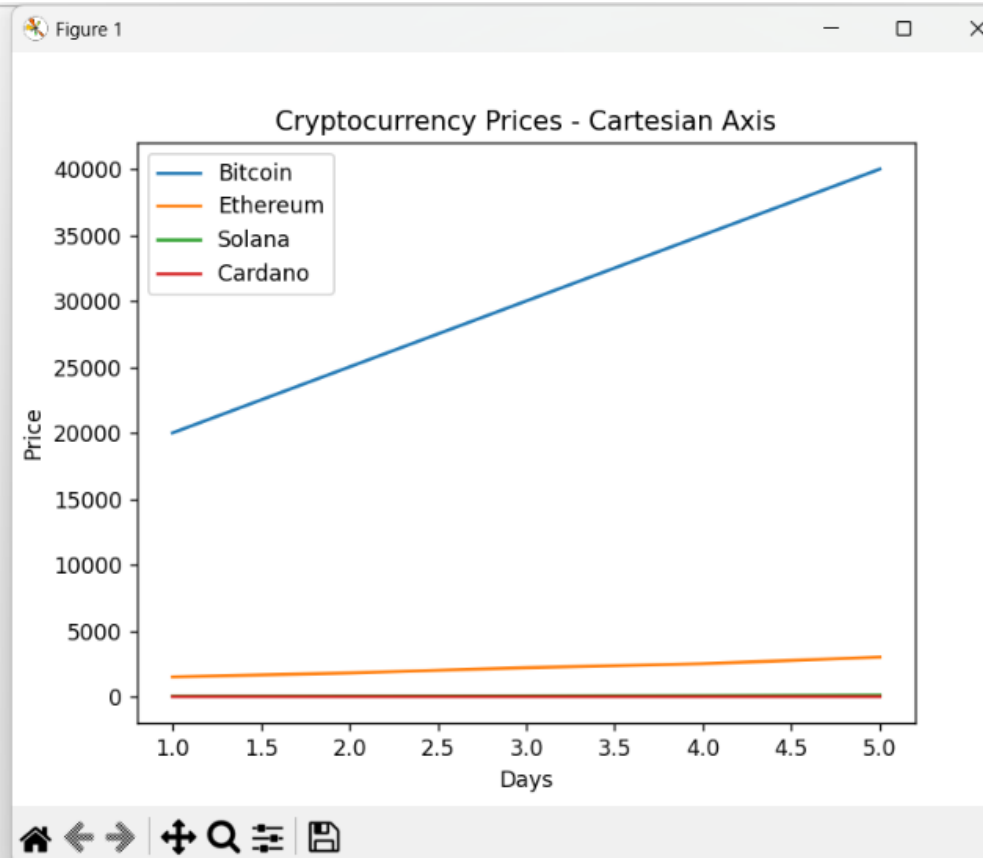
```
plt.plot(
    crypto["Day"],
    crypto["Solana"],
    label="Solana"
)
```

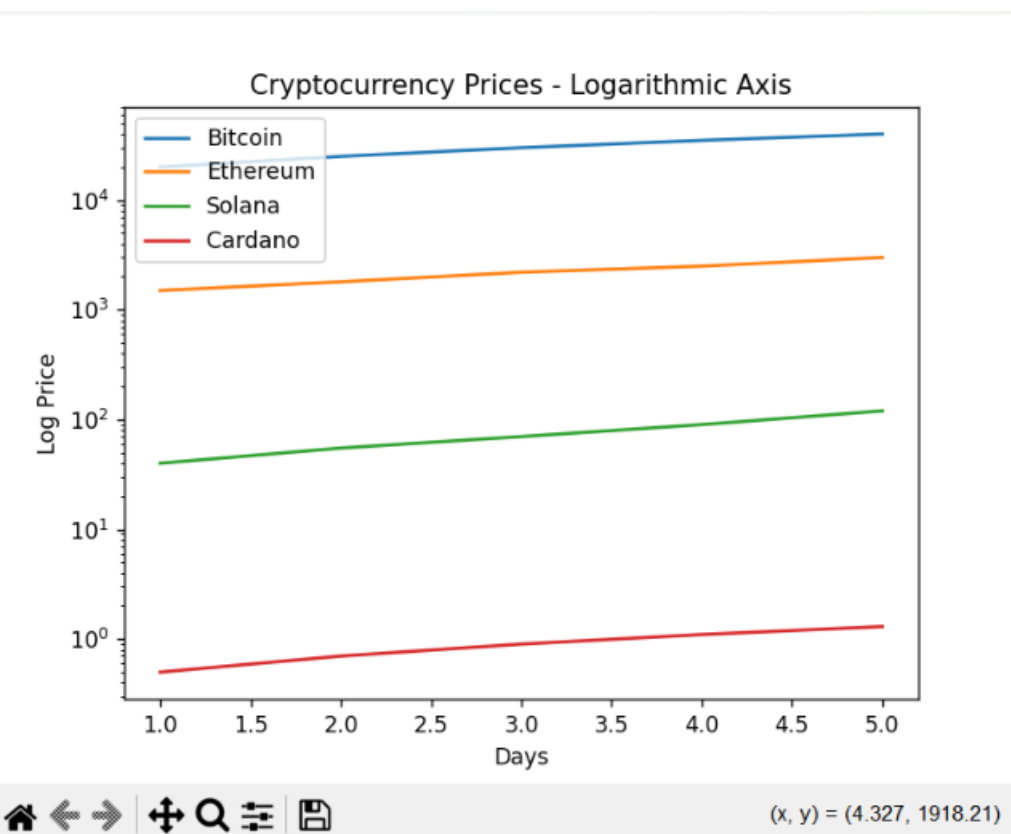
```
plt.plot(
    crypto["Day"],
    crypto["Cardano"],
    label="Cardano"
)
```

```
plt.title("Cryptocurrency Prices - Cartesian Axis")
```

```
plt.xlabel("Days")
plt.ylabel("Price")
```

```
plt.legend()
```





9 – Marathon Runner Performance Analysis

```
import pandas as pd

import matplotlib.pyplot as plt
import numpy as np
runner = pd.DataFrame({
    "Direction": [0, 45, 90, 135, 180],
    "Distance": [5, 10, 15, 20, 25],
    "Speed": [10, 12, 11, 9, 8]
})

# Cartesian Visualization
plt.scatter(
    runner["Distance"],
    runner["Speed"],
    c=runner["Speed"],
    cmap="plasma",
    s=200

plt.colorbar(label="Speed")
plt.title("Runner Speed Analysis")
plt.xlabel("Distance Covered")
```

```
plt.ylabel("Speed")
plt.show()
# Polar Visualizatio
theta=np.radians(runner["Direction"])
plt.figure(figsize=(6,6))
ax=plt.subplot(
    111,
    polar=True

scatter=ax.scatter(
    theta,
    runner["Distance"],
    c=runner["Speed"],
    cmap="viridis",
    s=200

plt.colorbar(scatter,label="Speed")
plt.title("Runner Direction Movement")
plt.show()
print("Cartesian plot is better for speed comparison.")
print("Polar plot provides better directional movement insights.")
```

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
```

```
runner = pd.DataFrame({
    "Direction": [0, 45, 90, 135, 180],
    "Distance": [5, 10, 15, 20, 25],
    "Speed": [10, 12, 11, 9, 8]
})
```

```
# Cartesian Visualization
```

```
plt.scatter(
    runner["Distance"],
    runner["Speed"],
    c=runner["Speed"],
    cmap="plasma",
    s=200
)

plt.colorbar(label="Speed")

plt.title("Runner Speed Analysis")

plt.xlabel("Distance Covered")

plt.ylabel("Speed")

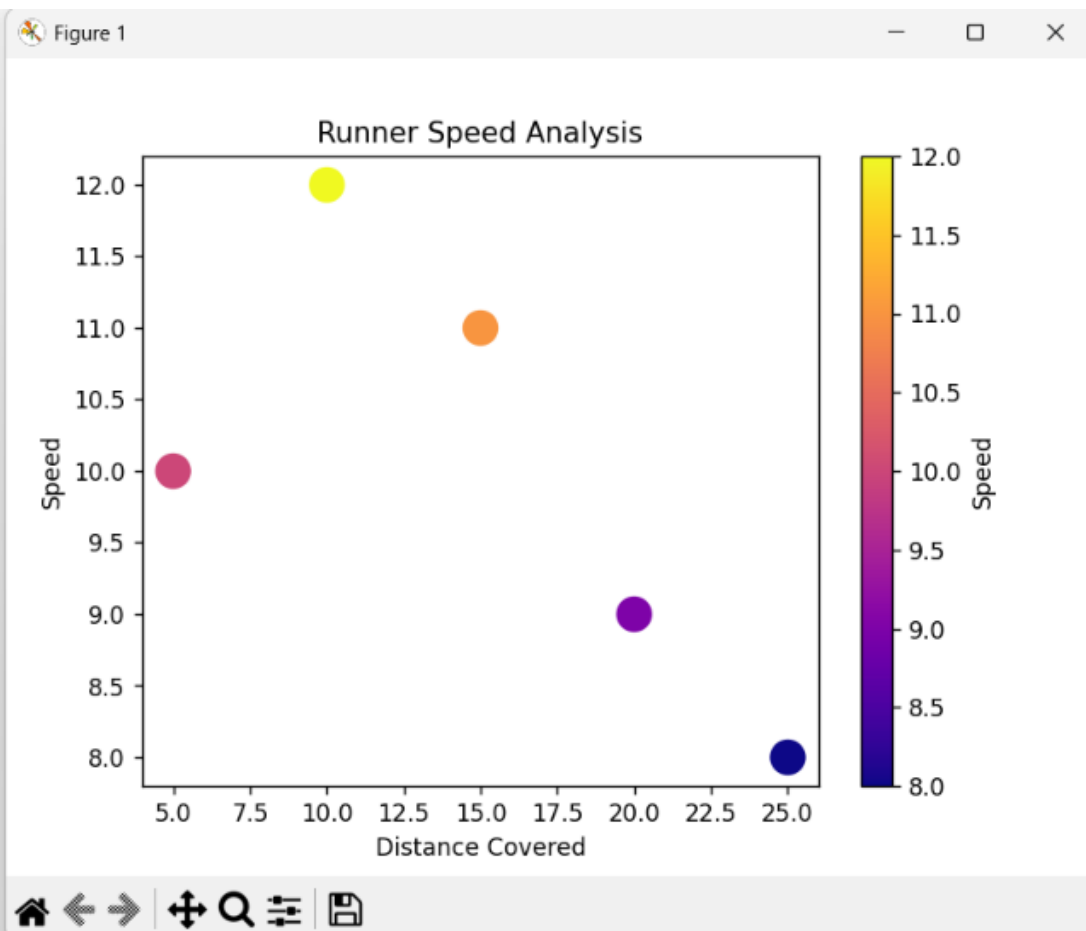
plt.show()
```

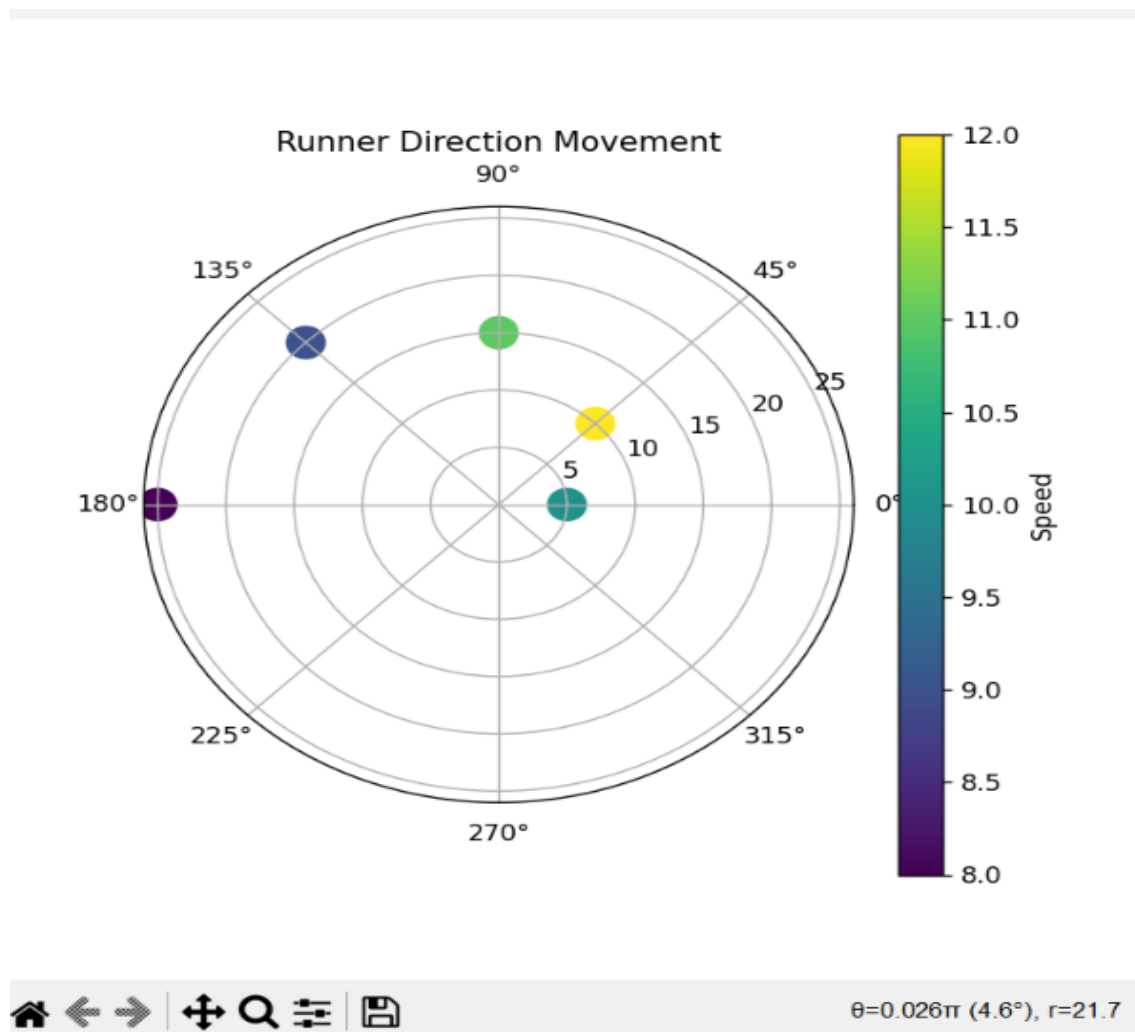
```
# Polar Visualization
```

```
theta=np.radians(runner["Direction"])
```

```
plt.figure(figsize=(6,6))
```

```
ax=plt.subplot(
    111,
    polar=True
)
```





10 – COVID-19 Vaccination Campaign Dashboard

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
vaccine = pd.DataFrame({
    "District":["Chennai","Madurai","Salem","Trichy","Coimbatore"],
    "Vaccination":[85,70,90,75,80]
})
# Palette 1
plt.bar(
    vaccine["District"],
    vaccine["Vaccination"],
    color=plt.cm.Blues(
        vaccine["Vaccination"]/100
    )
)
```

```
plt.title("Vaccination Rate - Blue Palette")
plt.xlabel("District")
plt.ylabel("Vaccination Percentage")
plt.xticks(rotation=45)
```

```
plt.show()
```

```
# Palette 2
```

```
plt.bar(
    vaccine["District"],
    vaccine["Vaccination"],
    color=plt.cm.Set2(
        range(5)
    )
)
```

```
plt.title("Vaccination Rate - Categorical Palette")
```

```
plt.xlabel("District")
```

```
plt.ylabel("Vaccination Percentage")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

```
print("Blue palette is better for policymakers because it shows intensity clearly.")
```

```
print("Categorical colors are better for public dashboards.")
```

```
print("Avoid red-green combinations for color vision deficiency users.")
```

```
import pandas as pd
import matplotlib.pyplot as plt

vaccine = pd.DataFrame({
    "District": ["Chennai", "Madurai", "Salem", "Trichy", "Coimbatore"],
    "Vaccination": [85, 70, 90, 75, 80]
})

# Palette 1
plt.bar(
    vaccine["District"],
    vaccine["Vaccination"],
    color=plt.cm.Blues(
        vaccine["Vaccination"]/100
    )
)

plt.title("Vaccination Rate - Blue Palette")
plt.xlabel("District")
plt.ylabel("Vaccination Percentage")
plt.xticks(rotation=45)
plt.show()

# Palette 2
plt.bar(
    vaccine["District"],
    vaccine["Vaccination"],
    color=plt.cm.Set2(
        range(5)
    )
)

plt.title("Vaccination Rate - Categorical Palette")
```

